

Statistical Screening for Poisoned Machine-Learning Training Data

Detection and Quarantine of Suspicious Samples at the Data-Ingestion Boundary

Evidence status: Refreshed against current code snapshot ed38ad039a60a648c38a247e4136b34bbe287bda and successful CI run 36783091096 on 2026-09-30.

Contribution

An ingestion-boundary screening pipeline combining statistical outlier methods with a label-aware cross-class centroid detector that improves the committed synthetic label-flip benchmark from the old top-1 spectral baseline 0.08 / 0.23 / 0.3675 F1 to 0.55 / 0.595 / 0.6975 F1 at 5% / 10% / 20% contamination.

Method

1. Validate incoming feature batches and streaming samples.
2. Apply z-score, IQR, Isolation Forest, and spectral/label-aware detectors.
3. Combine detector signals for screening and quarantine workflows.
4. Maintain streaming state and deduplication metadata.
5. Benchmark label-flip behavior with committed reproducible artifacts.

Current Verified Evidence

Current-main Python 3.12 CI reports:

- 162 tests passed, 0 skipped.
- 70.77% statement coverage; CI gate is 45%.
- Lint/format, Pyright, dependency audit, CodeQL, and the shipped-implementation benchmark smoke job succeeded.
- Committed results/spectral_benchmark.json records cross-class centroid F1:
 - 0.55 at 5% contamination
 - 0.595 at 10%
 - 0.6975 at 20%
- The same artifact retains the weaker top-1 baseline 0.08 / 0.23 / 0.3675.
- Cross-class spectral detection beats the feature-space ensemble on 3/3 tested contamination rates.

Reproducibility

Expected CI evidence: 162 passed, 70.77% coverage.

Honest Boundaries

- The benchmark is synthetic and separable; it is not proof of production detection rate.
- Clean-label, subtle backdoor, and very-low-rate poisoning remain difficult.
- Throughput is hardware/environment dependent; no fixed product SLA is claimed.
- Stateful streaming behavior requires deliberate partitioning/external state for horizontal scaling.